GAUSSIAN MOMENTS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

August 3, 2026

We reformulate results in Goessmann (2021) based on the formalism in Goessmann et al. (2026).

1 Gaussian chaoses

Let $g_k [X_k]$ be for $k \in [d]$ an independent vector of indepedent Gaussian coordinates. We build the tensor product

$$g_{\mathrm{EL}} \left[X_{[d]}\right] \coloneqq \langle \{g_k [X_k] \,:\, k \in [d]\} \rangle_{\left[X_{[d]}\right]} \,.$$

A gaussian chaos of order d is a contraction of this tensor product with a tensor $\tau \left[X_{[d]}\right]$.

We are interested in the moments of gaussian chaoses and relate them to properties in the tensor $\tau \left[X_{[d]}\right]$.

Definition 1 (J-injective norm)**.** *Let J be a partition of $[d]$ and $\tau \left[X_{[d]}\right]$ a tensor. The J-injective norm of $\tau \left[X_{[d]}\right]$ is then*

$$\|\tau \left[X_{[d]}\right] \|_J = \max \left(\left\langle \{\tau \left[X_{[d]}\right]\} \cup \{\theta^M[X_M] \,:\, M \in J\} \right\rangle_{[\varnothing]} \,:\, \left\langle \{\theta^M[X_M] \,:\, M \in J\} \right\rangle_{[\varnothing]} = 1 \right) .$$

Theorem 1 (Latala bound)**.** *There are to each d constants C_d and c_d such that for any $\tau \left[X_{[d]}\right]$ and $q \geq 1$*

$$c_d \sum_{J \subset [d]} \|\tau \left[X_{[d]}\right] \|_J \cdot q^{|J|} \leq \| \left\langle \tau \left[X_{[d]}\right], g_{\mathrm{EL}} \left[X_{[d]}\right] \right\rangle_{[\varnothing]} \|_q \leq C_d \sum_{J \subset [d]} \|\tau \left[X_{[d]}\right] \|_J \cdot q^{|J|}$$

2 Gaussian tensor networks

While gaussian chaoses are defined by tensor products, let us define general Gaussian tensor networks on hypergraphs $\mathcal{G}$ by tensors

$$g_{\mathcal{G}} [X_{\mathcal{V}}]$$

with coordinates

$$g_{\mathcal{G}} [X_{\mathcal{V}} = x_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} g_e [X_e = x_e] \,.$$

Especially we are interested on moment bounds on

$$\left\langle \tau \left[X_{[d]}\right], g_{\mathcal{G}} [X_{\mathcal{V}}] \right\rangle_{[\varnothing]}$$

where $[d] \subset \mathcal{V}$. Intuitively, averaging over hidden variables improves the concentration behavior of Gaussian tensor networks, due to the law of large numbers. Here we need to show effects of averaging over multiple hidden variables.

2.1 Reduction to Gaussian chaoses

The trick we apply is that any contracted Gaussian tensor network can be related to a Gaussian chaos on a mutilated hypergraph. This mutilation involves introduction of additional edges decorated by delta tensors.

To be more precise we define a new hypergraph

$$\mathcal{V} \cup \bigcup_{e\in\mathcal{E}} \bigcup_{v\in e} \{(e,v)\}$$

and edges

$$\{\{(e,v)\,:\, v\in e\}\,:\, e\in\mathcal{E}\} \cup \{\{(e,v)\,:\, v\in e\} \cup \{v\}\,:\, v\in\mathcal{V}\}$$

where the first set of edges is decorated as before and the second by delta tensors. Then

$$\left\langle \tau\left[X_{[d]}\right], g_{\mathcal{G}}\left[X_{\mathcal{V}}\right]\right\rangle_{[\varnothing]} = \left\langle \{\tau\left[X_{[d]}\right]\} \cup \{\delta[(X_{e,v} : v\in e), X_v]\} \cup \{g_e\left[X_{e,e}\right]\,:\, e\in\mathcal{E}\}\right\rangle_{[\varnothing]}$$

We notice that this is a Gaussian chaos to the structure tensor

$$\left\langle \{\tau\left[X_{[d]}\right]\} \cup \{\delta[(X_{(e,v)} : v\in e), X_v]\}\right\rangle_{\left[X_{(v,e)\,:\, e\in\mathcal{E}, v\in e}\right]}\,.$$

To show moment bounds, we are left to calculate the J-injective norms of this structure tensor.

2.2 Properties of J-injective norms

- Tensor product property: Multiplicativity of J-injective norms into those to factors.
- Delta tensor property (more general: ODECO).

3 Orlicz tensor networks

If the tensor network consists of i.i.d. tensors of i.i.d. Orlicz random variables, we can relate the moments of the contraction to the moments of a Gaussian chaos. In the dominating Gaussian chaos, $\alpha/2$ random tensors replace a single α Orlicz tensor.

4 Applications in Knowledge Graphs

Setup:

- Atom extraction queries: Treated as deterministic
- Importance query: I.i.d. bernoulli coordinates (i.e. 2-Orlicz). Rescaled for isotropy.

Aim:

- Predict the average of specific functions. This is the expected contraction with the isotropic importance query.

4.1 Concentration of single expectations

We first show concentration of the expectation to single formulas Bernoulli coordinates are bounded and therefore sub-Gaussian, that is 2-Orlicz. We can therefore apply the results on Orlicz tensor networks to show moment bounds.

4.2 Uniform concentration of multiple expectations

Having moment bounds for single expectations, we use chaining with Mendelsons Γ functionals to show uniform concentrations. We can then show probabilistic guarantee recovery on least squares regression.

References

Alex Goessmann. *Uniform Concentration of Tensor and Neural Networks: An Approach towards Recovery Guarantees*. PhD Thesis, Technische Universität Berlin, Berlin, 2021. URL https://depositonce.tu-berlin.de/handle/11303/15990.

Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL http://arxiv.org/abs/2601.15442. arXiv:2601.15442 [cs].